Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/06/2016
PI Name:	Robbins, Robert M.D.		
Project Title:	Vascular Health in Space: MicroRNAs in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Houston	**State:**	TX
Zip Code:	77021-2039	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	04/30/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 4/30/2017 as PI changed due to Dr. Robbins' move to University of Arizona. Per F. Hernandez/ARC; also note that start date changed by NSSC to 11/01/2014 and original start date was 10/28/2014 (Ed., 7/12/17)		
Key Personnel Changes/Previous PI:	NOTE (JULY 2017): PI change to Dr. Doris Taylor due to Dr. Robbins' move to University of Arizona and project will continue as "Vascular Health in Space: MicroRNAs in Microgravity (PI: Taylor)" with Dr. Doris Taylor as the new PI.		
COI Name (Institution):	Schrepfer, Sonja M.D., Ph.D. (University Medical Center Hamburg, Germany/University of California San Francisco (UCSF)) Schwartz, Robert Ph.D. (Texas Heart Institute) Taylor, Doris Ph.D. (Texas Heart Institute)		
Grant/Contract No.:	NNX15AB55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts are faced with several health risks during both short- and long-duration spaceflight due to the space environment. Microgravity itself seems to affect cardiovascular parameters. For example, studies on animals (rats and rhesus monkeys) have found consistent decreases in heart rate during spaceflight. Similarly, data from 12 astronauts over 6 missions showed that heart rates of astronauts were reduced in space. These early studies demonstrate an obvious change in an easy to measure cardiovascular parameter. Although cardiovascular disease is the leading cause of death in the United States, to our knowledge, cardiovascular physiology in space has not been studied, yet. More in-depth studies are now required to shed light on how microgravity exerts changes on individual cells and tissues. We believe that subcellular effects of microgravity can be documented and we hypothesize that microRNAs (miRNAs), because of their regulatory function, will be involved. miRNAs are now understood to be powerful modulators of biological pathways and have generated considerable excitement as next-generation therapeutic targets for a variety of diseases, including cardiovascular diseases. In this proposal, we will assess the gravitational impact on the miRNA response to vascular injury and will evaluate the efficiency of vascular repair in space. This will shed light on how microgravity alters vascular remodeling and healing and could reveal currently unknown targets for potential therapeutic modulation. We will evaluate the role of miRNAs in vascular disease (Objective 1), vascular health (Objective 2), vascular smooth muscle cell function (Objective 3), and angiogenesis (Objective 4). Previously, we investigated miRNA regulation in diseased human vessels with myointimal lesions and demonstrated the important regulatory role of miRNAs in vessel injury. Our studies demonstrated that various vascular miRNAs were downregulated after vessel injury and only a few were upregulated. Recent spaceflight studies showed that the expression of miRNAs in both men and mice were sensitive to gravity. The limited data available demonstrated down-regulation of multiple miRNAs in microgravity. We therefore hypothesize that microgravity could aggravate pathologic vessel remodeling via miRNA alterations. Because inflammation is a known stimulus for myointimal hyperplasia, we will pay special attention to the inflammatory responses to vessel injury. In Objective 1, we will investigate how vascular miRNAs are differentially regulated at 1xg and µg in response to vascular balloon injury. In Objective 2, we will investigate the effect of microgravity on miRNAs in healthy vessels. If the gravitational effect on vascular miRNAs is profound, microgravity might even induce the development of vascular lesions in healthy vessels. In Objective 3, we will assess the effect of microgravity on smooth muscle cell (SMC) function in both healthy and diseased vessels. Under normal physiologic conditions, SMCs rarely proliferate. However, in the presence of injury or a growth stimulus, SMCs begin to proliferate and migrate. Microgravity could be another so far unknown growth stimulus. Uncontrolled SMC proliferation can result in pathologic vasoconstriction or in myointimal hyperplasia with narrowing of the lumen. Microgravity might thus affect SMC plasticity. In Objective 4, we will assess and compare angiogenesis at 1xg and µg and define the role of angiogenetic miRNAs. Since angiogenesis plays an important role in vascular health, we anticipate that effects of microgravity on angiogenesis-associated miRNAs will impact vascular health and repair processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected gain in knowledge will also advance the research on vascular miRNAs on Earth. Our group and a few others have focused on altering miRNA expression in injured vessels using systemic or local delivery of anti-miRNAs. We believe that therapeutically altering miRNAs is a newly evolving strategy with enormous potential. The results we expect to achieve from this study will thus complement the work that has been done under static gravitation on Earth. They could provide insights into genetic regulation and signaling pathways and reveal new targets for therapeutic modulation, including potential new pharmaceutical products.
Task Progress:	The project team participated in a preliminary discussion with NASA staff, and since then has been standing-by for follow-up instructions for the planned live animal return studies. There might be an upcoming flight opportunity (SpaceX 12) and the team participated in telecons. In the meantime, the team submitted a data request proposal to NASA for the opportunity to investigate the vessels of the RR1 (n=10) and RR3 mice (n=10). Both requests have been approved and we first completed studies regarding tissue quality and its feasibility for histopathological and microarray/RNA sequencing analysis (data have been submitted to GeneLab). We are now analyzing the vessels of 10 flight carcasses, 10 ground control carcasses, 10 vivarium control carcasses, and 10 basal carcasses for both the RR1 and RR3 missions. For both missions, mice were frozen on the International Space Station (ISS). Therefore, we are not able to perform physiological or cell culture studies. RR1 carcasses were already thawed and frozen once before the team attended the dissection and therefore, the interpretation of the RR3 data is important. To study the mechanism of microgravity, the team also performed in vitro assays using human smooth muscle cells and the Synthecon rotator (developed by NASA). We are currently preparing 2 manuscripts deriving from our NASA studies and we are aiming to submit them in early 2017. We are also aiming to submit the data as abstracts to the ASGSR Annual Meeting in 2017. The team appreciates the productive and supportive collaboration with NASA and we look forward to continuing our studies with NASA. EDITOR'S NOTE (JULY 2017): Project continues as "Vascular Health in Space: MicroRNAs in Microgravity (PI: Taylor)" with Dr. Doris Taylor as the new PI; see that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated:)